LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	43
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need and components of FDS
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need and components of FDS.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of FDS block diagram and component interconnections
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the Flight Director System?
(b) Why is an AFCS necessary in modern aircraft?
(c) Can you list three basic components of a Flight Director?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of FDS • Need for automated flight guidance • Basic FDS block diagram overview • Role of AFCS in flight control • Integration with autopilot systems	L & Ds	PPT, IB	15 min
(b)	• Pitch command generation logic • Roll command generation logic • Sensor inputs for FDS • Error signal processing • Display symbology interpretation	L & Ds	PPT, PDF	20 min
(c)	• FDS failure modes and indications • Pilot override procedures • Maintenance access points • Troubleshooting common faults • Safety implications of FDS failure	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS uses sensor data to generate flight commands.
(b) Describe the pilot's role when the FDS indicates a system fault.
(c) Differentiate between the functions of the FDS and the autopilot.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from manual to automatic flight using FDS guidance, referencing LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief explanation of about typical ADI, HSI and flight computer. Brief explanation of EFIS and concept of symbology.	Refer to the master syllabus reference list	LP 44

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min